PBL Phase 3: Project Development, Prototype, and Exhibition

Preparation Purpose of Phase 3

Phase 3 focuses on moving from research and planning into **project development, prototype creation, and exhibition preparation**.

At this stage, students should begin showing clearly:

- What they are going to create
- How their research supports their idea
- What materials, equipment, space, or technology they need
- How their project will work
- How the project will be presented at the exhibition
- What visitors will learn from the project

The final exhibition should not consist only of written research. Each project should develop into a **model, prototype, website, experiment, business proposal, interactive demonstration, or another clear exhibition product**.

This work can later support the actual exhibition demonstration if appropriate materials and equipment become available.

Amber + Irene + Lyla + Belle – Team Company Project Team Goal

Amber, Irene, Lyla, and Belle are now developing **one proposed semiconductor/AI company in Taiwan**, inspired by their previous research, including Winstek.

You are no longer developing four completely separate

projects. Each person is responsible for one major part of the

company:

- **Amber:** Overall business model and investment plan
- **Irene:** Problems, risks, and solutions
- **Lyla:** AI business branch
- **Belle:** Training and education business branch

All four parts must eventually connect into **one complete company**

proposal. Your team should be able to explain:

1. What the company does
2. What products and services it provides
3. Who its customers are
4. How the company earns money

5. What equipment it needs
6. What facilities/buildings it needs
7. What employees/workers it needs
8. What technology it needs
9. What suppliers and partners it needs
10. What initial investment may be required
11. What problems and risks the company may face
12. How the company will solve or reduce those problems
13. How AI creates additional business and revenue
14. How training develops skilled workers and generates additional income 15. Why an investor should consider investing in the company 16. How the company can grow in the future

For the exhibition, combine all four sections into **one company display/model**.

Visitors should be able to follow the company from its services and operations through its costs, risks, AI services, training programs, customers, and final revenue/business results.

The team should eventually be able to present the company to an investor by explaining:

"This is our semiconductor/AI company. This is what we will do, how the company works, how we will earn money, what investment we need, what risks we face, how we plan to solve those problems, and why the business can grow."

Amber – Business Model and Investment Plan

You are responsible for developing the **overall business model** of the proposed company.

Use your previous Winstek research as a real-world reference, but remember that the final business should be **your team's own proposed semiconductor/AI company**.

Step 1 – Define what the company sells

Identify:

- Products
- Services
- Semiconductor-related services
- Other business services

Step 2 – Identify customers

Explain:

- Who the customers are
- Why they need the company's services
- How customers will pay the company

Step 3 – Identify revenue

Explain the company's major:

- Revenue sources
- Methods of generating income

Step 4 – Identify expenses and investment requirements

Research:

- Main expenses
- Equipment
- Facilities/buildings
- Employees
- Technology
- Suppliers
- Partners
- Initial investment requirements

Step 5 – Explain how the company operates

Explain how the company's different sections connect:

- Semiconductor services
- AI branch
- Training/education branch
- General company operations/management

Step 6 – Explain customer development and future

growth Research:

- How the company will attract customers
- Why customers would choose the company
- Future growth opportunities

Step 7 – Prepare a business-flow diagram

Create a simple diagram showing the connection between:

Customers → Services → Company Operations → Costs/Investment → Revenue/Profit

Also show how the AI and training branches connect to the main

company. **Step 8 – Prepare the investment case**

Explain:

- How much initial investment may be needed
- What the investment will be used for
- Why an investor should consider investing
- How the company could operate successfully and generate profit

Why this is important

Amber's work explains to visitors and investors **how the entire company works and makes money**.

Your research will form the main **Business Model and Investment section** of the team's exhibition model and investment presentation.

Lyla – AI Business Branch

You are responsible for designing the **AI department/branch** of the proposed semiconductor company.

Your main question is:

How can AI become a product or service that generates additional profit for the company?

Step 1 – Research possible AI services

Research how AI can help semiconductor companies or become a service that other companies may pay for.

Possible areas include:

- AI chip inspection
- AI defect detection
- Predictive equipment maintenance
- Semiconductor data analysis
- Production optimization
- AI customer support
- Semiconductor training AI
- Other useful AI services

Step 2 – Select the strongest ideas

For each selected AI service, explain:

What it does → Who needs it → What problem it solves → Why they need it → How it connects to the semiconductor company → How the company earns money from it

Step 3 – Identify resources

Explain what the AI department needs, including:

- Employees/workers
- Computers
- Software
- Data
- Technology
- Other necessary resources

Step 4 – Connect your branch to the team company

Work with Amber and Irene so that:

- Your AI business fits Amber's overall business model.
- Your AI risks and requirements connect with Irene's risk-and-solution work.

Step 5 – Develop a prototype if possible

Your priority is to create a **working version of your AI** and provide a usable link before adding unnecessary extra features.

If possible, develop a simple working AI prototype that visitors can test at the exhibition.

Why this is important

The AI branch demonstrates how the company can create:

- Additional revenue
- New products/services
- Future growth opportunities

Your AI branch should appear clearly as one part of the final company exhibition model.

Belle – Training and Education Business Branch

You are responsible for designing the company's **training and education**

department.

Use your previous research about TSMC training, but now develop it into a business service that can also generate revenue.

Step 1 – Identify the target students

The training program may serve:

- University students
- High-school students
- New semiconductor employees
- People preparing for semiconductor careers

Our training center is for:

1. **University students**
    - Learn basic semiconductor knowledge.
    - Get hands-on experience.
2. **High school students**
    - Learn about semiconductors.
    - Explore future career choices.
3. **New semiconductor employees**
    - Learn about equipment, processes, and safety.
    - Get ready for their jobs.
4. **The people who want to work in the semiconductor industry**
    - Learn the skills they need.
    - Prepare for a semiconductor career.

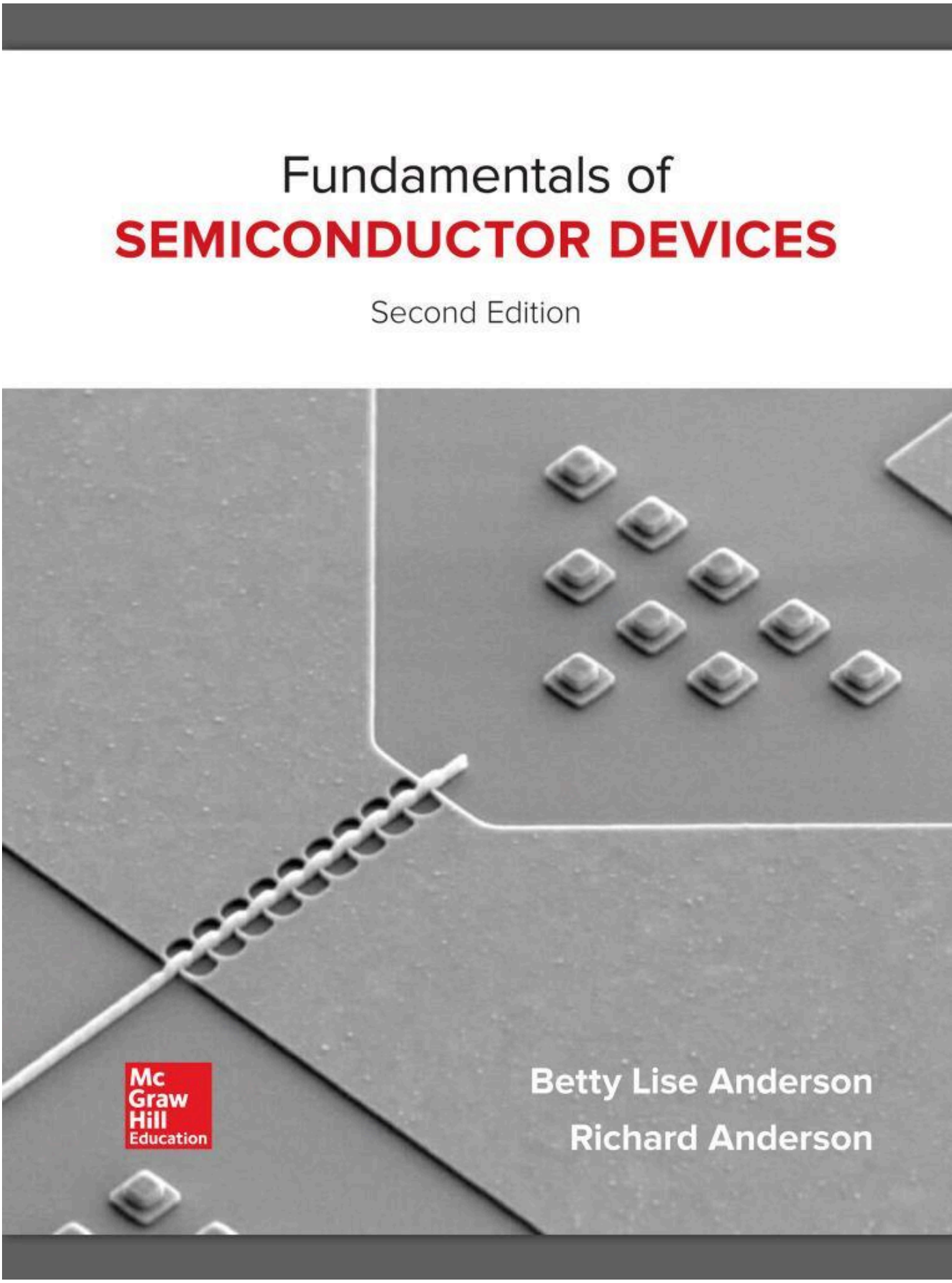
Step 2 – Decide what courses will be provided

Possible courses include:

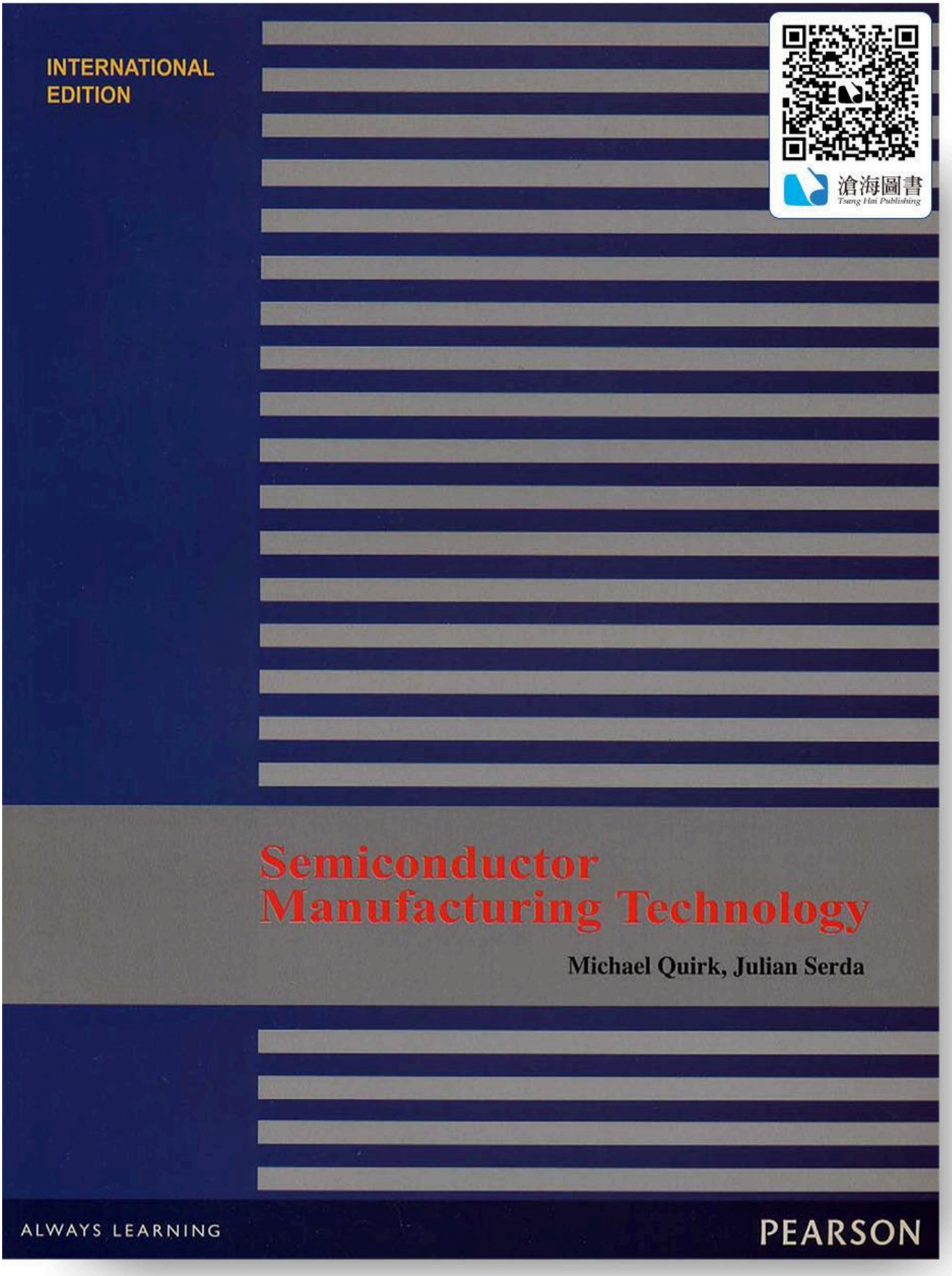
- Semiconductor basics
- Equipment training
- Process training
- Cleanroom and safety
- AI/intelligent manufacturing
- Hands-on semiconductor activities

Our training center has these courses

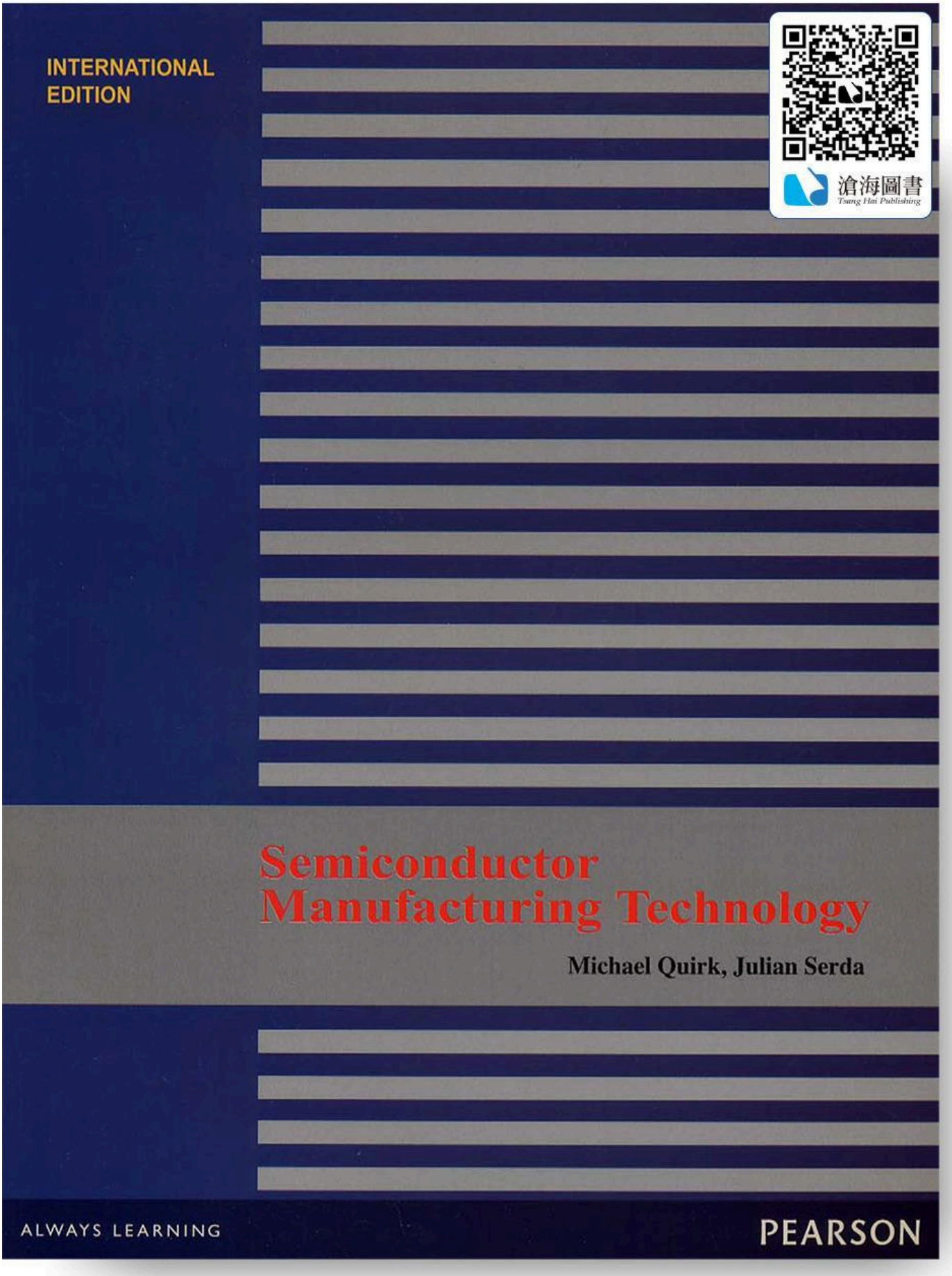
1. **Semiconductor Basics** — *Fundamentals of Semiconductor Devices*



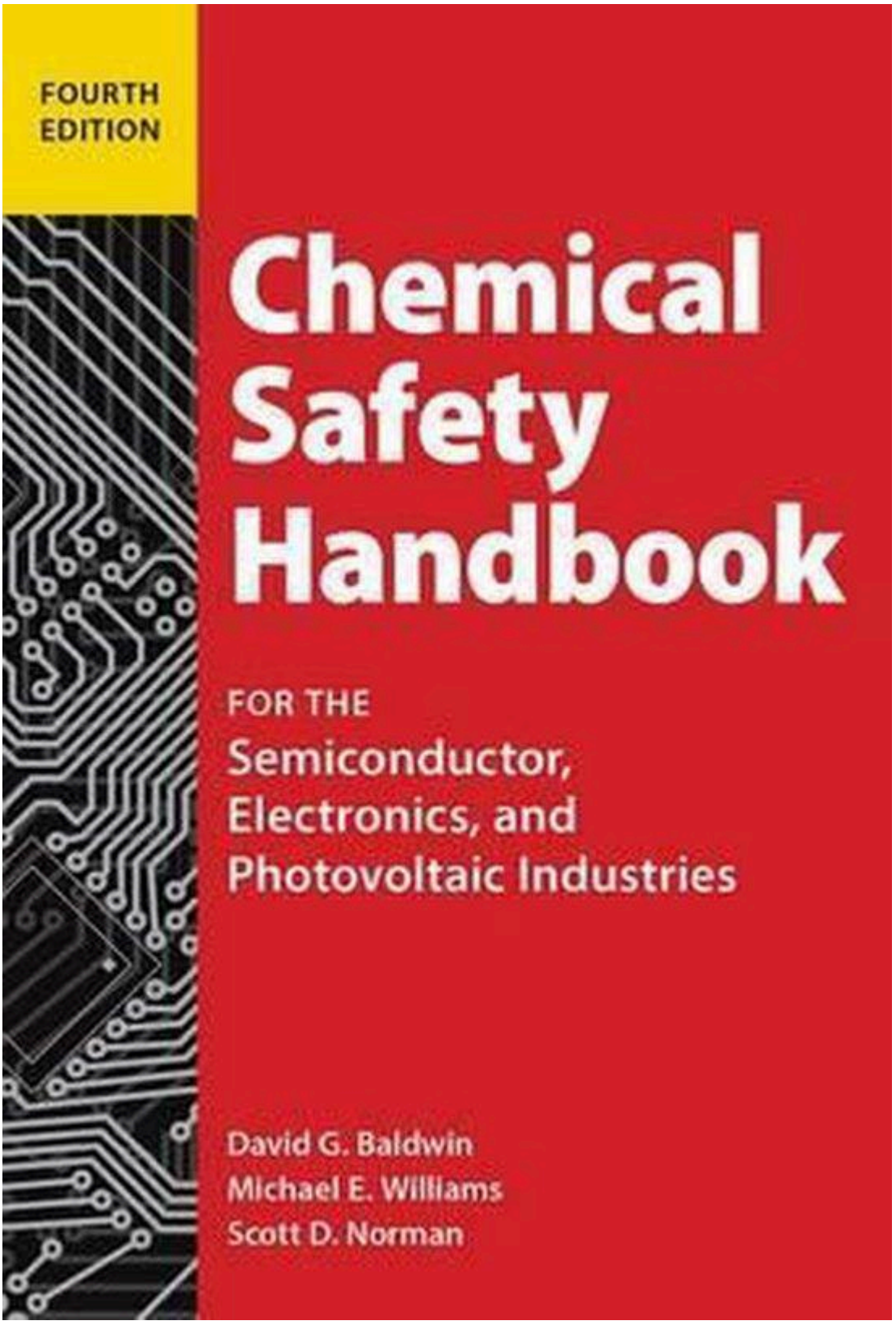
2. **Semiconductor Process** — *Semiconductor Manufacturing Technology*



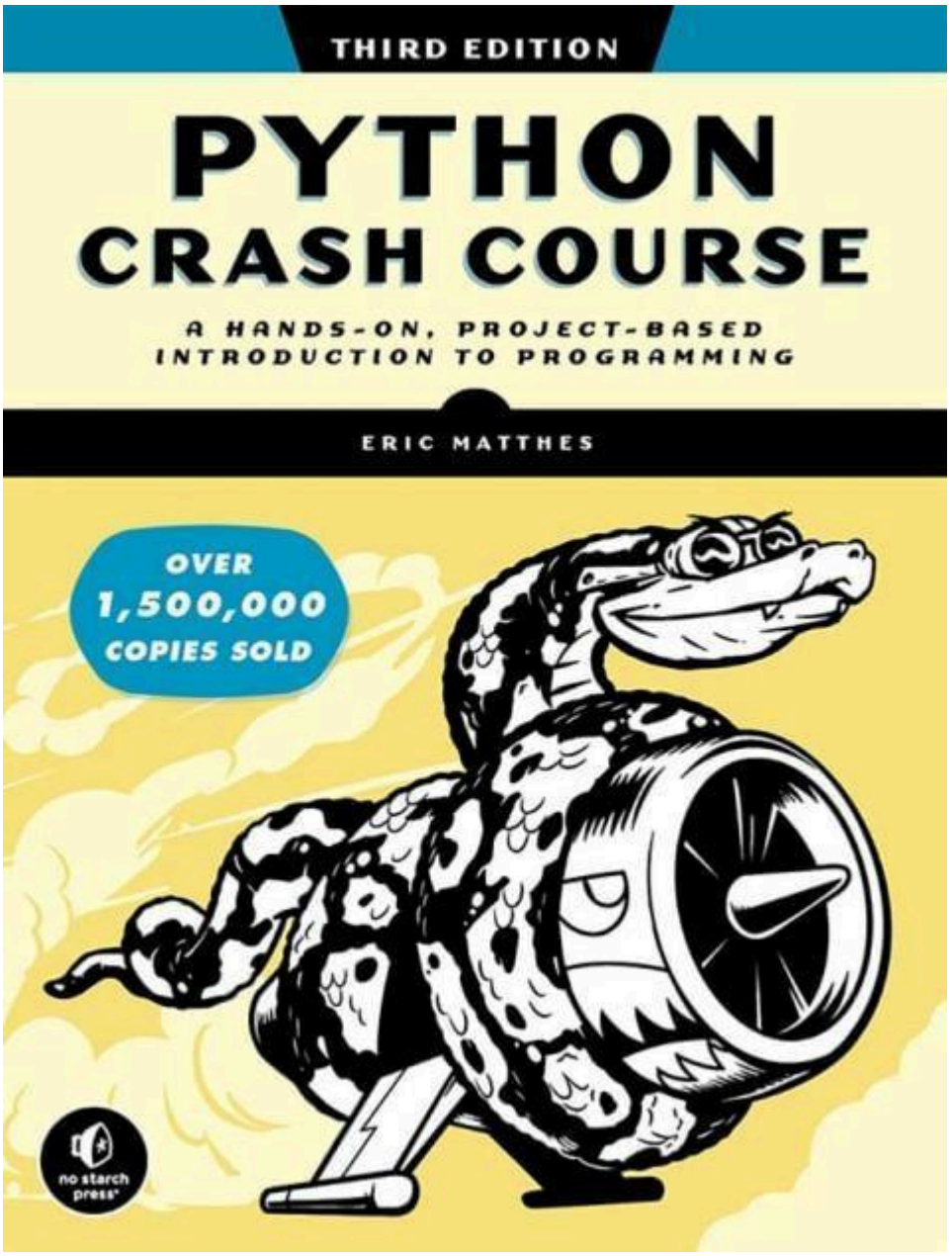
3. **Equipment Training** — *Semiconductor Manufacturing Technology*



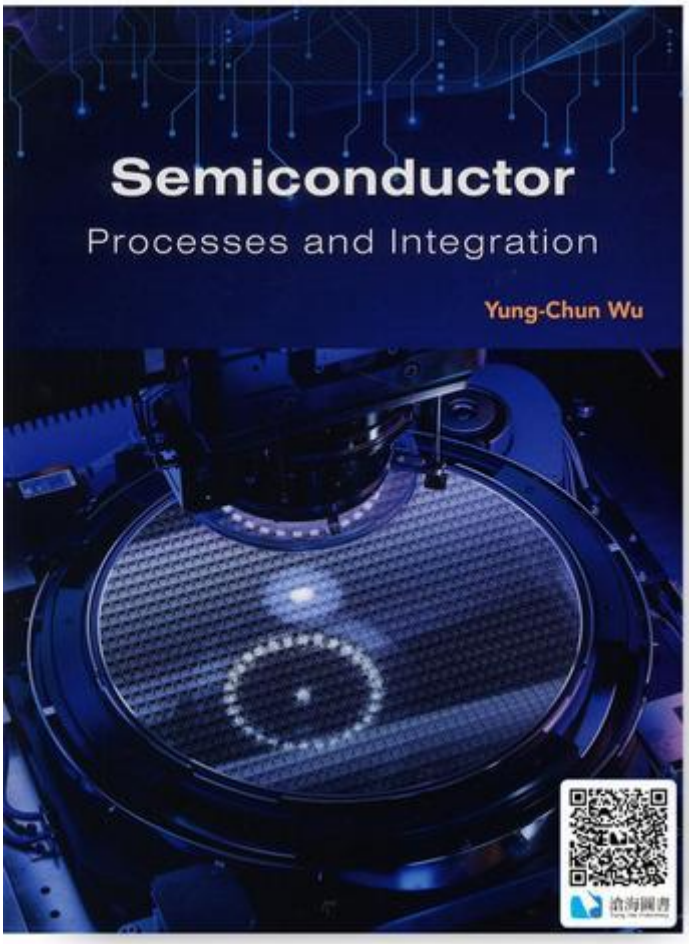
4. **Cleanroom and Safety** — *Semiconductor Safety Handbook*



5. **AI and Programming** — *Python Crash Course*



6. **Hands-on Practice** — *Semiconductor Processes and Integration*



Students can choose classes based on their level:

- **Basic:** Basics and Safety
- **Intermediate:** Process and Equipment
- **Advanced:** AI, Programming, and Practice

Step 3 – Plan the training spaces

Decide exactly what rooms/classrooms/training areas are

required. For each room, identify:

- Course/training taking place there
- Approximate room size
- Number of students it can hold
- Important equipment required

Student Training Plan

Step 1: Register

Students sign up for courses.

Step 2: Check Level
Teachers check students' levels.

Step 3: Learn Basics
Students learn basic semiconductor knowledge.

Step 4: Learn Safety
Students learn safety rules.

Step 5: Learn Skills
Students learn processes, equipment, AI, and programming.

Step 6: Practice
Students practice with equipment.

Step 7: Take a Test
Teachers check students' skills.

Step 8: Get a Certificate
Students who pass get a certificate.

1.Classroom A	Semiconductor Basics	60 m²	20	Desks, chairs, whiteboard, projector
2.classroom B	Semiconductor Process	60 m²	20	Desks, chairs, whiteboard, projector
4.Computer / AI Lab	Computer, AI, and programming	65 m²	15	Computers, projector
5.Equipment Training Room	Equipment training	80 m²	15	Training equipment, tools
6.Process Training Lab	Process practice	100 m²	15	Process equipment, tools
7.Cleanroom Training Room	Cleanroom and safety	80 m²	15	Cleanroom equipment, safety equipment

8.Hands-on Lab	Wafer and testing	80 m²	15	Wafers, testing equipment
9.Equipment Room	Equipment practice	60 m²	15	Semiconductor equipment, tools
10.VR / Safety Room	Safety practice	50 m²	15	VR equipment, safety equipment
13.Cafeteria / Coffee Area	Eat and drink	100 m²	30	Tables, chairs, refrigerators
14.Student Rest Area	Rest and table tennis	100 m²	20	Sofas, tables, table tennis table
15.Staff Area	Teacher and employee rest	40 m²	10	Tables, chairs, lockers
16.Office Area	Office work	50 m²	10	Desks, computers, printers
17.Meeting Room	Meetings	40 m²	15	Table, chairs, projector
18.Public Restrooms	For students and staff	140 m²	—	Toilets, sinks, hand dryers
19.Storage Area	Store things	30 m²	—	Shelves, cabinets
20.Security Area	Security	20 m²	2	Monitors, computers, cameras
21.Corridors and Stairs	Walking and moving	300 m²	—	Lights, signs, handrails
22.Elevator	Go to other floors	25 m²	—	Elevator, control panel

Area				
23.Electrical and Mechanical Area	Electricity and air-conditioning	110 m²	—	Electrical equipment, air-conditioning equipment
24.Cleanroom Air System Area	Cleanroom air system	40 m²	—	Air filters, air system
25.IT Area	Computers and network	20 m²	—	Servers, network equipment
26.Cleaning Area	Cleaning supplies	10 m²	—	Cleaning tools, shelves
27.Garbage and Recycling Area	Garbage and recycling	20 m²	—	Bins, recycling containers
28.Equipment Repair Area	Repair equipment	30 m²	—	Tools, workbench
29.Fire Control Area	Fire safety	15 m²	—	Fire alarm system, control panel
30.Loading Area	Move equipment	80 m²	—	Loading equipment, carts
31.Emergency Training Area	Safety training	100 m²	20	Fire extinguishers, safety equipment
32.Bus / Drop-off Area	School bus and student drop-off	150 m²	—	Bus stop, signs
33.Small Parking Area	Staff and visitor parking	100 m²	—	Parking signs
34.Outdoor Rest / Green Area	Rest and plants	60 m²	20	Benches, plants

Total Area

Indoor area: 2191m²

Outdoor area: 490 m²

Total site area: about 2,681 m²

People

About 120 people at one time

About 100 students

About 20 teachers and staff

Regular classes: **20 students**

Lab classes: **15 students**

Step 4 – Plan hands-on learning

Decide:

- What hands-on activities will be included
- What training equipment is needed
- What facilities students will use
- How students will gain practical experience

1. **Wafer practice**
    - Students practice using wafers.
2. **Equipment practice**
    - Students learn how to use semiconductor equipment.
3. **Process practice**
    - Students practice basic semiconductor processes.
4. **Cleanroom practice**
    - Students learn how to work safely in a cleanroom.
5. **Testing practice**
    - Students practice testing semiconductor products.
6. **AI and programming practice**
    - Students use computers to practice AI and programming.

Step 5 – Plan how the training business operates

Explain:

- Who can enroll
- How students will enroll
- How long courses will last
- How the training program operates

- How enrollment/course fees can generate income

Step 5: Business Operation

- **Students:** High school students, university students, and new workers.
- **Registration:** Students register online.
- **Test:** Students take a simple test.
- **Classes:** Students join a class based on their level.
- **Course time:** 1 week to 3 months.
- **Class size:** 15–20 students.
- **Money:** Students pay for the courses and dormitory.

Step 6 – Connect training to the company

Explain how the training center supports the company's need for skilled workers. **Submission**

- **Students learn useful skills.**
- **Students get practice.**
- **Students learn safety.**
- **Companies can find trained workers.**

Today: Upload details of the required rooms/classes, including:

- Course/training
- Approximate room size
- Student capacity
- Important equipment

This information will later be used to create the company's **training-center floor plan and model**.

Why this is important

The company needs skilled workers, but the training center can also operate as a separate **business service that generates revenue**.

Your design will form the **Training and Education section** of the final company model and exhibition presentation.

Irene – Company Problems, Risks, and Solutions

You are responsible for investigating:

What problems could this proposed semiconductor/AI company face, and how can the company prepare solutions before those problems happen?

Step 1 – Examine every major part of the company

Look for possible problems involving:

- Employees/talent
- Equipment failure
- Semiconductor manufacturing
- Quality control
- Electricity
- Water
- Materials
- Suppliers
- Production capacity
- Safety
- Environment
- Finance
- Customers
- Competition
- Supply chain
- Technology changes

Step 2 – Organize the risks

Group the problems into clear business areas such as:

- Workers
- Equipment
- Production
- Materials
- Electricity
- Water
- Finance
- Customers
- Competition
- Supply chain
- Technology
- Safety/environment

Step 3 – Explain every major problem using the same

structure For each problem, write:

Problem → Why/Cause → Effect on the Company → Solution/Backup

Plan Step 4 – Prioritize serious risks

Focus especially on problems that could:

- Stop company operations
- Interrupt production
- Reduce quality
- Lose customers
- Increase costs
- Cause the company to lose money

Step 5 – Coordinate with Amber

Discuss your risks and solutions with Amber so that the risks you investigate match the company's actual business model.

Why this is important

Investors need to understand not only how the company could make money, but also:

- What could go wrong
- How serious the risks are
- How the company plans to manage those risks

Your research will become the **Risk and Solution section** of the company's investment proposal and exhibition model/presentation.

Link:

https://esg.tsmc.com.tw/en-US/articles/271?utm_source

https://esg.tsmc.com.tw/file/public/e-AnAdmiredEmployer.pdf?utm_source

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